

No. of Functional Features Included in the Solution

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	5 Marks

Functional Features Overview:

Functional features included in the FitBuddy solution, with purpose, module, implementation status and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Input Form & Validation	Collects name, user ID, age, weight, goal, intensity and rejects invalid values	index.html, schemas.py	Done	0.5
2	AI 7-Day Plan Generation	Creates a personalised Day 1 to Day 7 plan using Gemini	gemini_generator.py, /generate-workout	Done	1
3	Nutrition / Recovery Tip	Gives a short, balanced tip for the chosen goal	gemini_flash_generator.py	Done	0.5
4	Feedback-Based Plan Update	Revises the whole plan from user feedback, building on the latest version	/submit-feedback	Done	1
5	Persistent Storage	Saves users, original plan, updated plan and tip	database.py (SQLite)	Done	0.5
6	Admin Dashboard	Lists and deletes users; password protection to be enforced	/view-all-users, /delete-user	In Progress	0.5
7	Retry & Error Handling	Retries Gemini up to 3 times; friendly error pages	gemini_client.py, main.py	Done	0.5
8	Health Check & Docker	Status endpoint and container image for deployment	/health, Dockerfile	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7 done, 1 in progress
Core / Must-Have Features	5 (features 1, 2, 4, 5, 7)
Additional / Nice-to-Have Features	3 (features 3, 6, 8)
Features Tested & Verified	[fill after running pytest]

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Input form, result page with feedback box, loading overlay, error and admin pages
2	Backend / Logic	3	Input validation, plan generation and update, nutrition tip
3	Database / Storage	1	Users table with plans and tip
4	API / Integration	1	Google Gemini API with retry and backoff
5	Security / Authentication	1	Admin HTTP Basic login, secrets in environment variables, input validation